

PROJECT REPORT

FINZEN

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List of APIs

1. MongoDB Atlas

- Used as the cloud database (via pymongo)
- Connection string format:
`mongodb+srv://<username>:<password>@cluster0.bjzstq0.mongodb.net/`

2. Machine Learning Models

- Custom-trained classifiers:
 - `type_classifier.pkl` (Wants/Needs)
 - `cat_classifier.pkl` (General category)
 - `needs_cat_classifier.pkl` (Needs-specific category)
 - Vectorizers:
 - `vectorizer.pkl` (General text)
 - `vectorizer_needs.pkl` (Needs-specific text)
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Assets

1. FinPet GIFs

- Paths: gifs/gif1.gif, gifs/gif2.gif, gifs/gif3.gif
- Used for pet evolution visuals (Ghastly → Haunter → Gengar)

2. Emoji Icons

- Used throughout the UI

3. Synthetic Dataset

- Hardcoded expense examples for model training:
 - 70 labeled entries across 7 categories (Food, Utilities, etc.)

Python Libraries

1. Core Libraries

- **os** - For operating system interactions
- **datetime** - For date/time handling
- **hashlib** - For password hashing (SHA-256)
- **pickle** - For model serialization/deserialization
- **urllib.parse** - For URL encoding (used in MongoDB connection)

2. Data Processing & ML

- **pandas** - For data manipulation
- **scikit-learn (as sklearn)** - For ML models:
 - **CountVectorizer** - Text feature extraction
 - **RandomForestClassifier** - Classification model
 - **train_test_split** - Data splitting for ML

3. Visualization

- **matplotlib.pyplot** - For static visualizations (pie charts)

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- `altair` - For interactive visualizations (bar charts)

4. Database

- `pymongo` - MongoDB Python driver
- `certifi` - For SSL certificate validation (MongoDB Atlas)

5. Streamlit

- `streamlit` - Core framework for the web app
- `st.experimental_rerun` / `st.rerun` - For UI updates

6. Image Handling

- `PIL.Image` - For image processing (FinPet GIFs)

Github repository Link

<https://github.com/manas1462005/FinZen>